# SIMATS ENGINEERING

## ASSESSMENT TOOL 2: Concept Mapping

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

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**COURSE OUTCOME COVERED**

**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

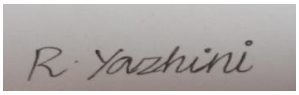
Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

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**Code of Conduct**

I R.YAZHINI(192521190) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



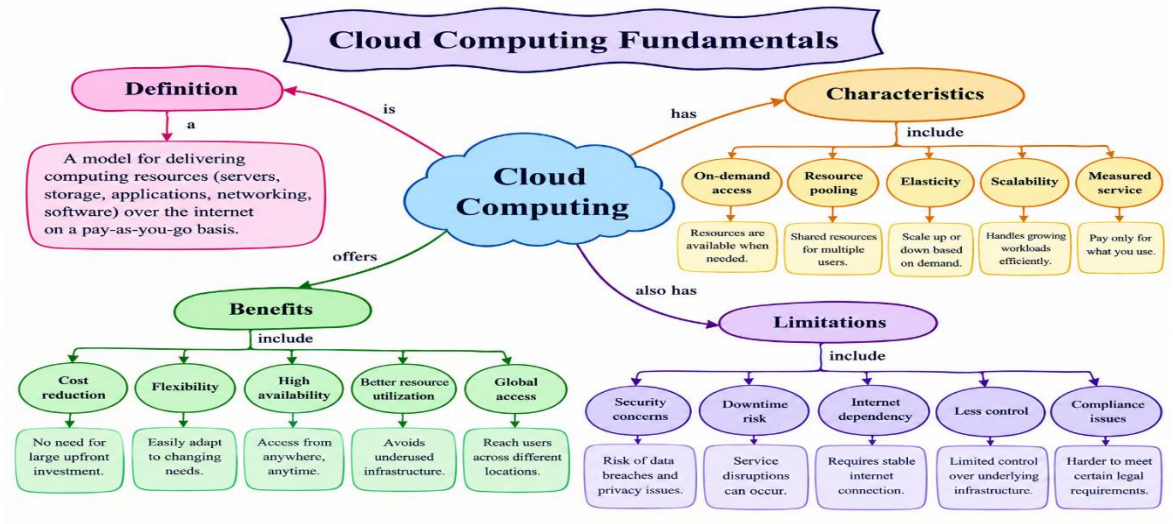
Signature of the student

**Assessment Tasks**

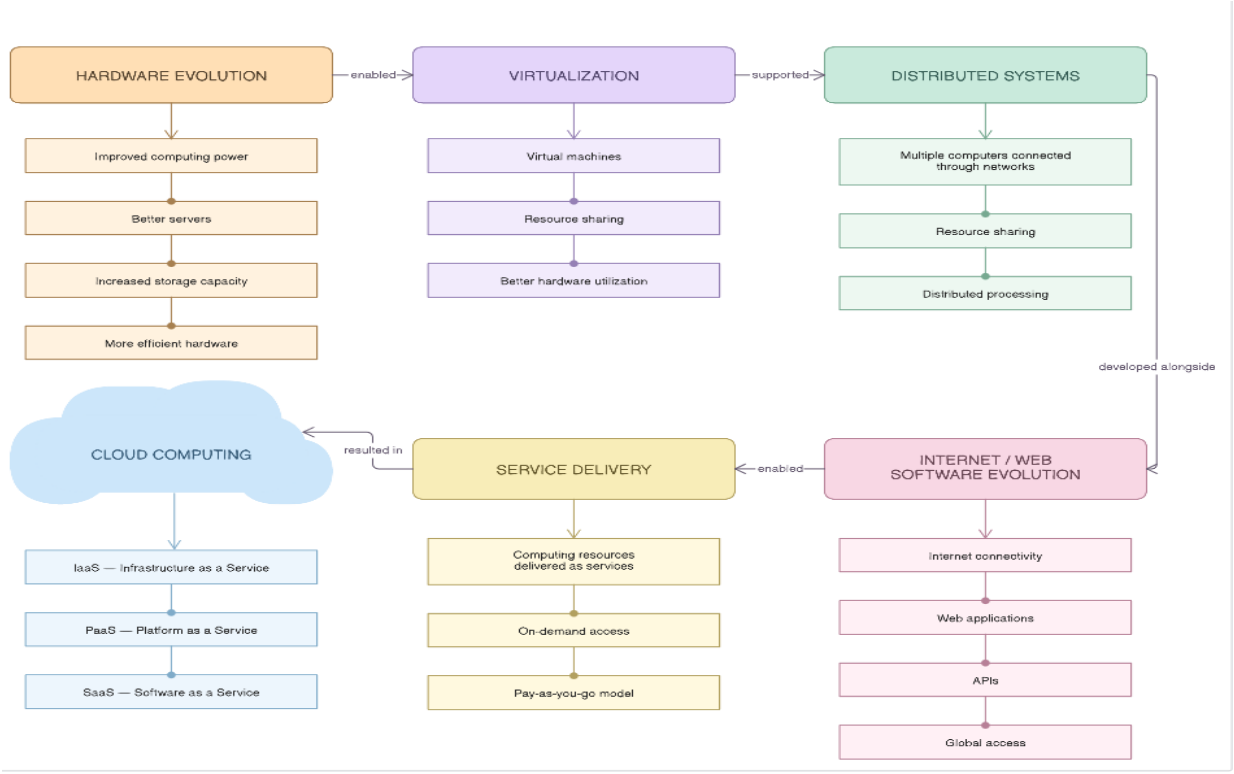
**Concept Mapping Tasks**

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

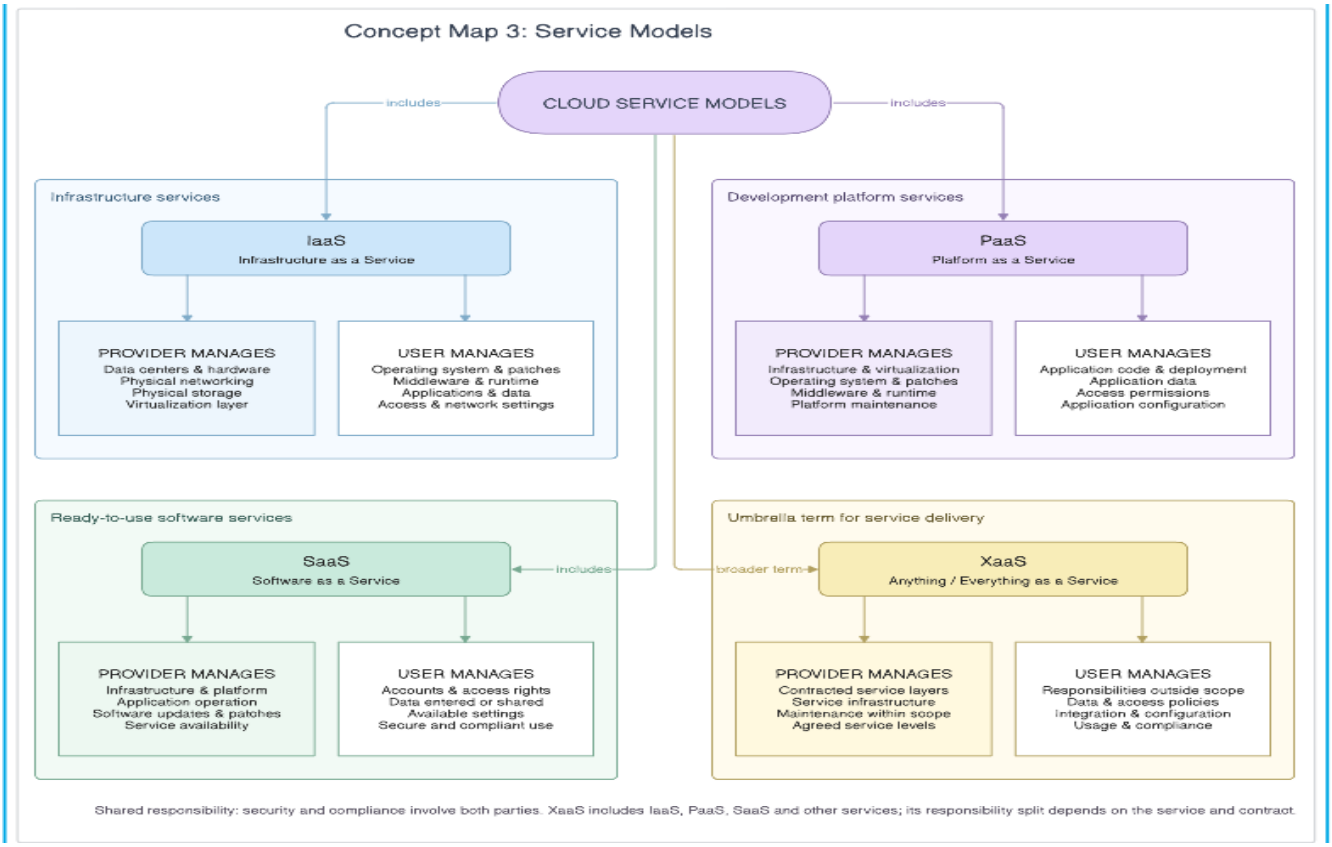
1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



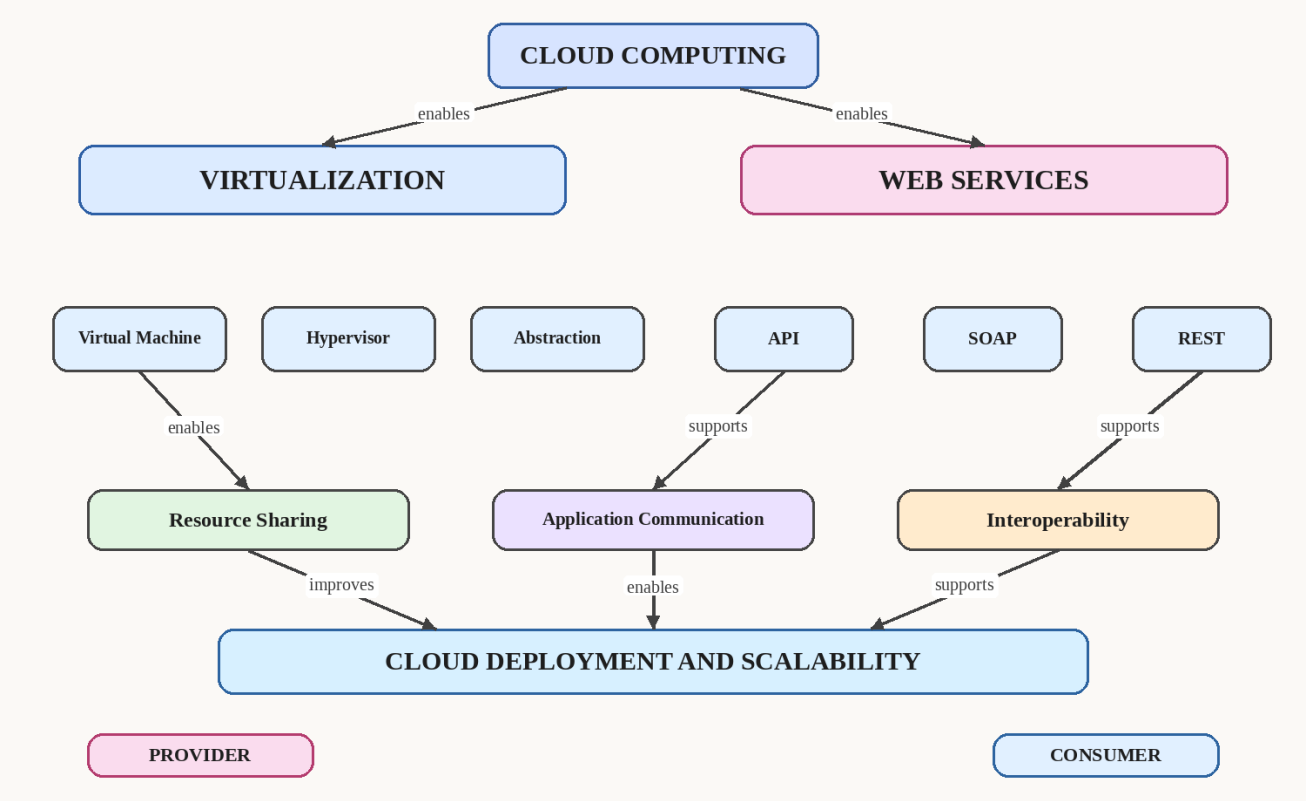
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.




3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.




**Expected concept nodes:**

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

**Concept Mapping Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Concept Identification | 25% | All major concepts are accurately identified | Most concepts are identified correctly | Limited concepts are identified | Essential concepts are missing |
| Relationship Mapping | 25% | Links between concepts are accurate and meaningful | Most links are correct | Some links are weak or unclear | Relationships are mostly incorrect |
| Hierarchy and Organization | 20% | Excellent structure with clear hierarchy | Mostly organized with minor issues | Basic structure with weak hierarchy | No logical organization |
| Technical Relevance | 20% | Map strongly reflects cloud course content | Mostly aligned to topic | Partially aligned to topic | Weak alignment to topic |
| Visual Clarity | 10% | Neat, readable, and easy to interpret | Generally readable | Somewhat cluttered | Poorly presented |